

Technics SX-WSA1R

System Exclusive Reference

Acoustic Modeling Synthesizer Module

Operating system version 2.0

Contents

1	Overview	2
1.1	Applicability	2
1.2	Ports	2
1.3	Conventions	2
2	Message framing	3
2.1	Identifiers	3
2.2	Termination	3
2.3	Length limit	3
3	The Technics message set	4
3.1	Message families	4
3.2	The model triple	4
3.3	Short messages	4
3.4	Parameter families	4
4	Bulk dump	5
4.1	Categories	5
4.2	Data message	5
4.3	Payload encoding	5
4.4	Checksum	6
4.5	Opening a transfer	6
4.6	Acknowledgement and retry	6
4.7	Continuation	6
4.8	Closing	6
4.9	Receiving	6
5	Universal System Exclusive	8
5.1	Recognised messages	8
5.2	Messages not recognised	8
6	Errors	9
6.1	The messages	9
6.2	ERROR 40 — wrong identification	9
6.3	ERROR 41 — reception	9
6.4	ERROR 42 — transmission	10
6.5	Interference from other traffic	10
7	Scope of this edition	11

Chapter 1

Overview

The Technics SX-WSA1R exchanges data over MIDI System Exclusive. This reference describes the messages it sends and accepts: their byte layout, the bulk-dump transfer protocol, the universal messages it honours, and what it reports when a transfer fails.

1.1 Applicability

This reference describes operating system version **2.0**. An earlier version 1 exists; its behaviour may differ and is not covered.

1.2 Ports

The instrument has two sets of MIDI sockets, MIDI 1 and MIDI 2. Ordinary performance data is handled on both. **Bulk dump transfers use MIDI 1 only** — a dump offered on MIDI 2 is not received, and the instrument sends dumps only from MIDI 1.

1.3 Conventions

Byte values are hexadecimal. A field written *nn* takes one of the values listed for it. Ellipses inside a message stand for a body whose length depends on the message.

Every message begins with **F0** and ends with **F7**, as required by MIDI.

Real-time messages are safe inside a transfer

System Real Time bytes (**F8–FF**) may be interleaved anywhere inside a System Exclusive message without disturbing it. Conversely, while the instrument is receiving a System Exclusive message it ignores **F8** TIMING CLOCK, **FA** START, **FB** CONTINUE and **FC** STOP for transport purposes.

Chapter 2

Message framing

2.1 Identifiers

The instrument accepts exactly two System Exclusive identifiers:

identifier	meaning
50	Matsushita (Technics)
7E	Universal Non-Real Time

A message carrying any other identifier is tracked to its F7 and discarded. No error is reported for it.

There is no device or unit number after the identifier

Many instruments place a device id immediately after the manufacturer identifier. The SX-WSA1R does not. The byte following 50 is already the message-family byte described in chapter 3. A unit field does appear later in some families, but it is not where the convention would put it.

2.2 Termination

A message ends at F7. If any other status byte in the range 80–F6 arrives while a message is open, the message is abandoned and its contents discarded.

2.3 Length limit

A message may not exceed 254 bytes. A longer message is rejected and, inside a bulk-dump session, reported as a reception error.

Chapter 3

The Technics message set

3.1 Message families

After F0 50, the third byte selects a message family. The instrument accepts fourteen values:

21 22 23 24 25 27 28 29 2A 2B 2C 2D 7E 7F

26 is not accepted. The SX-WSA1R's own product identifier is 2D; its outgoing bulk-dump data messages all begin F0 50 2D.

3.2 The model triple

The families 21, 22, 2B, 2C and 2D continue with three fixed bytes:

04 *nn* 11

where *nn* is 00 or 01. This triple identifies the product; a message whose triple does not match is rejected with ERROR 40.

3.3 Short messages

These messages are complete as shown. They form the control vocabulary of a bulk-dump session.

message	meaning
F0 50 23 7E F7	acknowledge
F0 50 24 7E F7	not acknowledged
F0 50 27 7E F7	end of category
F0 50 28 7E F7	end of dump
F0 50 29 7E F7	abort
F0 50 2A 7E F7	destination full
F0 50 21 04 <i>nn</i> 11 F7	enquiry
F0 50 22 04 <i>nn</i> 11 F7	enquiry reply

3.4 Parameter families

The families 2B and 2C carry individual parameter traffic rather than bulk transfer. Each addresses a large parameter space, and neither is enumerated in this edition; see chapter 7.

Chapter 4

Bulk dump

Bulk dump transfers the instrument's internal data as System Exclusive. The instrument is silent for the duration, and the transfer uses the MIDI 1 sockets only.

4.1 Categories

The SYSEX BULK DUMP screen offers five categories:

TOTAL KEYBOARD
SOUND
COMBINATION
SYSTEM, PART & MIDI
SEQUENCER

4.2 Data message

F0 50 2D 04 *nn* 11 *a2 a1 a0 l2 l1 l0 data... c sum* F7

nn unit field, 00 or 01.

a2 a1 a0 destination address, three bytes, most significant first, seven bits each: the value is $a_2 \times 128^2 + a_1 \times 128 + a_0$.

l2 l1 l0 length in *source* bytes, encoded the same way.

data the payload, encoded as described below.

c continuation flag: 01 if further messages follow, 00 on the last.

sum checksum.

A message is filled to at most 252 bytes before being sent; longer transfers are split.

4.3 Payload encoding

Each source byte is sent as two message bytes, **high nibble first**:

$$b \longrightarrow (b \gg 4), \quad (b \& 0F)$$

So source byte F1 appears on the wire as 0F 01, and the payload occupies twice as many bytes as the length field states. A receiver rejoins them as $((\text{first} \& 0F) \ll 4) | (\text{second} \& 0F)$.

An odd number of payload bytes is invalid.

4.4 Checksum

Let S be the eight-bit sum of every byte from the identifier 50 through the continuation flag inclusive. Then

$$sum = (0 - S) \& 7F$$

The leading F0, the checksum byte itself and the trailing F7 take no part in the sum. Equivalently, the sum of every byte after F0 including the checksum is zero modulo 128.

The checksum is not verified for every family

On reception the checksum is checked only for families 7E, 2B, 2C and 2D. For the other accepted families a wrong checksum passes unremarked.

4.5 Opening a transfer

A dump begins with a two-step exchange. Each step is attempted up to three times.

1. The sender transmits F0 50 21 04 00 11 F7 and expects F0 50 22 04 *nn* 11 F7.
2. The sender transmits F0 50 22 04 00 11 F7 and expects F0 50 23 7E F7.

If the exchange succeeds, every subsequent data message must be acknowledged. If all three attempts of the first step fail, **the dump proceeds anyway, without acknowledgements**, at a fixed pace of roughly 51 ms between messages.

4.6 Acknowledgement and retry

With acknowledgements in force, each data message must be answered with F0 50 23 7E F7. Any other reply causes the same message to be retransmitted, up to three attempts, after which the transfer is abandoned.

A reply is awaited for about 5.1 seconds; during the opening exchange the wait is about 2.05 seconds.

4.7 Continuation

Data messages after the first open with F0 50 7E, followed by payload, continuation flag, checksum and F7.

4.8 Closing

At the end of each category the sender transmits F0 50 27 7E F7 and waits for an acknowledgement. At the end of the whole transfer it transmits F0 50 28 7E F7. If the transfer was abandoned after three failed retries it transmits F0 50 29 7E F7.

4.9 Receiving

When the SX-WSA1R is the receiver it answers each data message with F0 50 23 7E F7 to accept, F0 50 24 7E F7 to reject, and F0 50 2A 7E F7 when the incoming data is larger than the destination.

While a bulk-dump session is open the instrument accepts only the 50 identifier; universal messages are not honoured until the session ends.

Chapter 5

Universal System Exclusive

5.1 Recognised messages

The instrument honours two universal messages, and transmits both.

message	meaning
F0 7E 7F 09 01 F7	General MIDI System On
F0 7E 7F 09 02 F7	General MIDI System Off

Turning General MIDI on restricts the sounds and operations available and may rearrange the percussion map. General MIDI System Off has no effect if General MIDI is already off.

Both messages are also written into Standard MIDI Files produced by the instrument, as a five-byte System Exclusive event at delta time zero.

5.2 Messages not recognised

A controller expecting any of the following will receive no response:

message	
F0 7F 7F 04 01 <i>ll mm</i> F7	Master Volume
F0 7E <i>dd</i> 06 01 F7	Identity Request
F0 7E 7F 09 03 F7	General MIDI 2 System On

The instrument does not implement Identity Reply, so it cannot be auto-detected by a controller that relies on it. The entire Universal Real Time identifier 7F is unrecognised.

Chapter 6

Errors

Transfer failures are reported on the SYSEX BULK DUMP screen. A System Exclusive message arriving while any other screen is displayed produces no visible error.

6.1 The messages

screen	meaning
COMPLETED!	the transfer finished
ERROR 40!	the message is addressed to a different product
ERROR 41!	reception failed
ERROR 42!	transmission failed
ERROR 21!	the destination is full

Not every failure is a 40, 41 or 42

When incoming data is larger than the destination area, the instrument reports ERROR 21! MEMORY FULL. Software that recognises only the three documented System Exclusive errors will mis-read a truncated restore as something else.

6.2 ERROR 40 — wrong identification

Raised when the message family byte is accepted but the bytes that follow it do not match the product. For a message addressed to the SX-WSA1R this means the model triple was not 04, then 00 or 01, then 11.

6.3 ERROR 41 — reception

The general reception failure. It covers:

- a message inside a session that does not begin F0 50;
- a status byte other than F7 arriving inside an open message;
- a framing, parity or overrun error on the MIDI input, or loss of active sensing;
- no reply within the timeout;
- a message longer than 254 bytes;

- an odd number of payload bytes;
- a continuation flag that is neither 00 nor 01;
- a checksum mismatch.

6.4 ERROR 42 — transmission

Raised when the instrument cannot complete an outgoing transfer internally, roughly a second after the attempt stalls. In this version it has a single cause and does not indicate a fault on the MIDI cable or at the receiving device.

6.5 Interference from other traffic

While a bulk-dump session is open, unrelated System Exclusive traffic arriving on the MIDI input can set the error state, and will then be reported when the session ends. Keep other sources off the MIDI 1 input for the duration of a transfer.

Chapter 7

Scope of this edition

This reference covers message framing, the bulk-dump transfer protocol, the universal messages, and error reporting. The following are not covered.

Parameter addresses. The message families 2B and 2C carry individual parameter traffic and each addresses a large parameter space. The addresses themselves are not tabulated here.

Bulk-dump data layout. The protocol that carries a dump is described in full, but the meaning of the bytes inside a dumped block — the structure of a sound, a combination or a sequence — is not.

Operating system version 1. Everything here describes version 2.0.